

Name:

Enrolment No:



MANIPAL UNIVERSITY
JAIPUR

Odd Semester End Term/Make Up Examination, January 2025

Faculty of Engineering
Department of Chemistry
B.Tech. Ist Year

Course Code: CHY1001/ CY1001

Course: Engineering Chemistry

Semester: I

SOLUTION

SECTION A (Memory Based Questions)

S.No.		Marks	
Q 1	Which of the following is not a natural polymer? (a) Rayon (b) Starch (c) Cellulose (d) RNA Sol: (a) Rayon (2 marks)		
Q 2	Distinguish between Chemical corrosion and Electrochemical corrosion. Sol: Chemical Corrosion: Occurs due to a direct chemical reaction between a material and a non-electrolytic environment (e.g., dry gases). Example: Oxidation of metals in dry air. Electrochemical Corrosion: Involves a redox reaction with electron transfer in the presence of an electrolyte (e.g., water). Anodic and cathodic regions form. Example: Rusting of iron in moist air. (2 marks)		
Q 3	What are batteries? Give the types of batteries. Sol: A battery is a device that stores chemical energy and converts it into electrical energy through electrochemical reactions. (1 mark) Batteries are broadly classified into two main categories: 1. Primary Batteries (Non-rechargeable): Designed for single use and cannot be recharged once depleted. 2. Secondary Batteries (Rechargeable): Can be recharged and reused multiple times by reversing the chemical reaction. (1 mark)	2 x 5 = 10	
Q 4	The hardness of water is due to (a) K and Na salts (b) Salts of iron (c) Salts of Ca and Mg (d) SiO ₂ Sol: (c) Salts of Ca and Mg (2 marks)		
Q 5	Select a non-homogeneous system from the following. (a) Salt solution (b) Sugar solution (c) Glucose in water (d) Super-saturated solution of NaCl Sol: (d) Super-saturated solution of NaCl (2 marks)		

SECTION B (Concept Based Questions)

Q 6	50 ml of standard hard water containing 1 gram of pure CaCO_3 per liter consumed 20 ml of EDTA. 50 ml of hard water consumed 25 ml of same EDTA solution and EBT indicator. Calculate the total hardness of water sample in ppm. Sol: Total hardness = 1250 ppm (6 marks) Can give step marks for calculation of molarity of $\text{CaCO}_3 = 0.01 \text{ M}$ (1 mark), EDTA = 0.025 M (1 mark), Molarity of hard water = 0.0125 M (1 mark)		
Q 7	What is hydrogen-oxygen fuel cell? Discuss its working. Sol: It is one of the most successful fuel cell. The cell consists of two inert porous electrodes made of graphite impregnated with finely divided 'Pt' (or) Ni (or) Pd – Ag alloy and a solution of 2.5% KOH as electrolyte. (1 mark) H_2 & O_2 gases are bubbled through anode & cathode compartments respectively. The following reactions takes place. Cell Reaction: At anode: $2\text{H}_2 (\text{g}) + 4\text{OH}^- \rightarrow 4\text{H}_2\text{O} + 4\text{e}^-$ At Cathode: $\text{O}_2 (\text{g}) + 2\text{H}_2\text{O} + 4\text{e}^- \rightarrow 4\text{OH}^-$ Net Reaction: $2\text{H}_2 (\text{g}) + \text{O}_2 (\text{g}) \rightarrow 2\text{H}_2\text{O}$, $E_{\text{cell}} = 1.23 \text{ V}$ Diagram (3 marks), description (2 marks)	6 x 5 = 30	
Q 8	Explain the mechanism of cationic polymerization reaction with suitable examples. Sol: Description (1 mark) with reactions (5 marks) taking any example		
Q 9	What is principle of Boy's calorimeter? Draw a schematic diagram of Boy's calorimeter. Sol: Principle: principle involved in this method is to burn the gas at a known constant rate in a vessel under such conditions that the entire amount of heat produced is absorbed by water which is also flowing at a constant rate. Thus the heat liberated by the fuel is equal to the heat absorbed by the water and the calorimeter. (2 marks)		

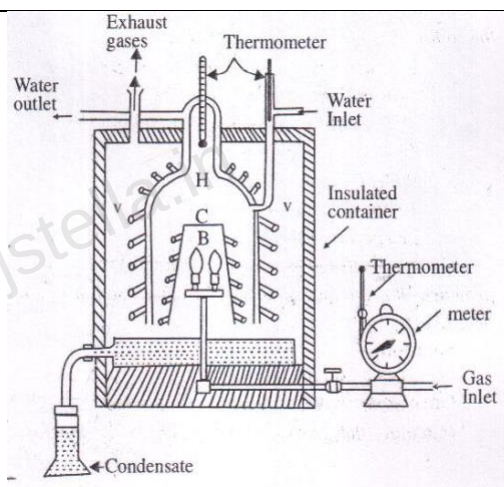
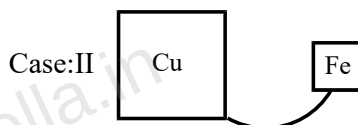
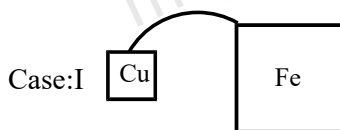


Diagram (4 marks)

Q 10

Study the following two cases. The metals are connected and in presence of some electrolyte. Which metal will corrode? In which case rate of corrosion will be faster? Justify your answer. (the area of the squares indicates the relative surface areas of the metals)



Sol: In both cases Fe (more active metal, anode) will corrode. Cu will act as cathode. (3 marks)

In case-II- the rate of corrosion will be faster, because the surface of cathode area is greater than anode area. Due to high corrosion current density on anode, rate of corrosion will be higher. (3 marks)

SECTION-C (Analytical Based Questions)

Q 11

Calculate the pilling Bedworth (P-B) ratio of the following metal oxides and find out which of the metal oxide coatings will be protective?

Metal	Metal Oxide	Atomic weight of metal	Metal oxide density (g/ cm ³)	Metal density (g/ cm ³)
Mg	MgO	24.3	3.65	1.74
Ca	CaO	40.08	3.40	1.55
Al	Al ₂ O ₃	27.00	3.70	2.70

Sol: (i) 0.79 (non-protective)

(ii) 0.638 (non-protective)

(iii) 1.37 (protective)

Each carry 3 marks

$$R_{PB} = \frac{V_{\text{oxide}}}{V_{\text{metal}}} = \frac{M_{\text{oxide}} \cdot \rho_{\text{metal}}}{n \times M_{\text{metal}} \cdot \rho_{\text{oxide}}}$$

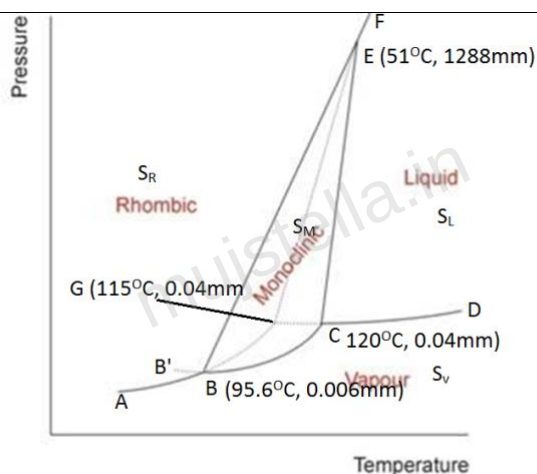
P-B ratio Formula (1 mark)

Q 12

Draw and explain phase diagram of sulfur. Find the degree of freedom for each equilibrium system.

Sol: Diagram (6 marks)

10 x 2
= 20



Schematic phase diagram for sulphur showing existence of rhombic and monoclinic allotropes
Dotted lines mark equilibria between metastable phases

Explanation (2 marks). Calculation of the degree of freedom for each equilibrium system (2 marks)

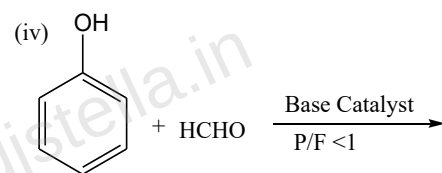
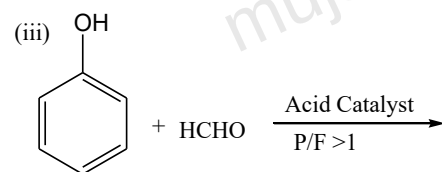
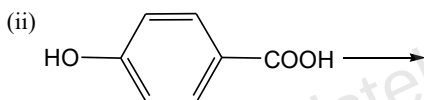
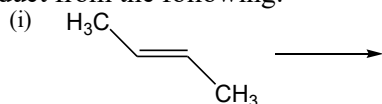
SECTION-D (Case Study/Application based Question)

Q 13 Calculate the water equivalent value of a bomb calorimeter using following experimental data:
Weight of benzoic acid = 2 g; weight of water= 2000 g; initial temperature= 25.0°C; final temperature = 30.0°C; fuse wire correction= 25 cal; cooling correction = 0.1°C. (calorific value of benzoic acid = 6325 cal/g)

Sol: water equivalent $w = 485.29$ gram (10 marks)

Only Formula (2 marks).

Q 14 Predict the polymer product from the following:



10 x 2
= 20

Sol: (each carry 2.5 marks)

